

Algebra II | Guided notes — we fill these in together

Vocabulary

Vertex Form: $y = a(x - h)^2 + k$, showing the vertex (h, k) directly.

Axis of Symmetry: The vertical line $x = h$ through the vertex.

Max/Min Value: The y-coordinate of the vertex, k , occurring at $x = h$.

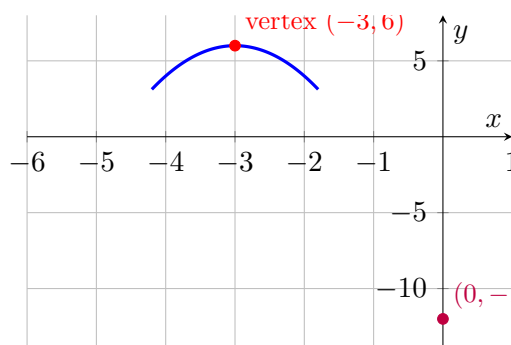
We do together. For $y = -2(x+3)^2 + 6$, find the vertex, y-intercept, transformations, and max/min value.

Step 1. Vertex: $a = -2$, $h = -3$, $k = 6$, so vertex = $(-3, 6)$.

Step 2. y-intercept: $y = -2(0 + 3)^2 + 6 = -2(9) + 6 = -12$, point $(0, -12)$.

Step 3. Transformations from $y = x^2$: shift left 3, reflect over x-axis, stretch narrower by 2, shift up 6.

Step 4. Since $a = -2 < 0$, this is a maximum value of 6, occurring at $x = -3$.



$y = -2(x + 3)^2 + 6$ showing vertex and y-intercept together.

You Try. Solve each. Show every step, then name the answer.

- a) For $y = (x - 1)^2 - 4$, find the vertex, y-intercept, and max/min value. Vertex $(1, -4)$, opens up;
y-intercept: $y = (0 - 1)^2 - 4 = -3$, point $(0, -3)$, mirror point $(2, -3)$; minimum value -4 at $x = 1$.

Summary (fill in): This week's four skills all connect back to one idea: the vertex (h, k) of $y = a(x - h)^2 + k$. From it you can graph the parabola, describe its transformations from $y = x^2$, and identify its maximum or minimum value.